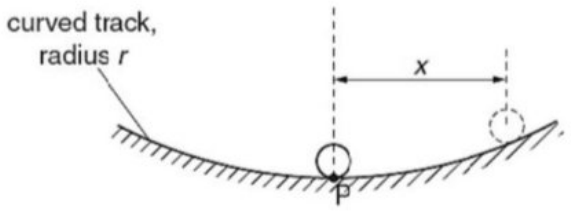
2 A small ball of mass 90 g rests at point P on a smooth curved track of radius r, as shown in Fig. 2.1.



The ball is moved a small distance to one side and is then released. The horizontal displacement x of the ball is related to its horizontal acceleration a towards P, by the expression

$$a = -\frac{gx}{r}$$

(a) Explain why the horizontal oscillation undertaken by the ball can be taken as simple harmonic motion.

.....

.....

.....[2]

(b) The radius r of curvature of the track is 28 cm.

Determine the angular frequency of the oscillation.

angular frequency = ... rad s^{-1} [2]

(c) After some time, the variation with time t of the displacement x of the ball is plotted as shown in Fig. 2.2. T is the period of oscillation.

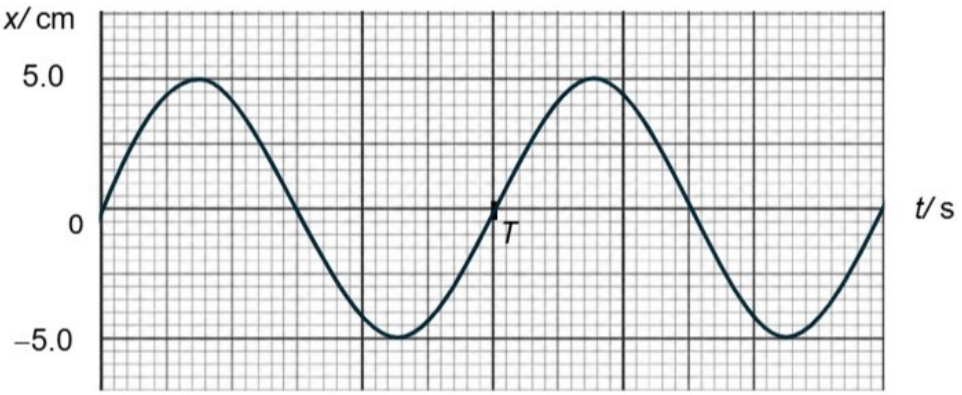


Fig. 2.2

(i) On the axes of Fig. 2.3, sketch the variation with time t of the acceleration a of the ball. [2]

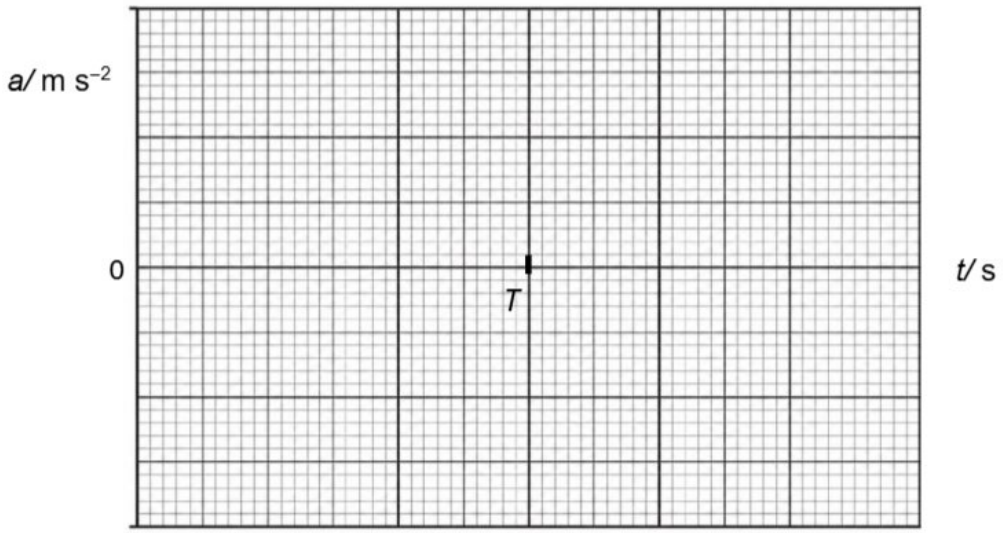


Fig. 2.3

(ii) The track is subsequently brought to a room with high humidity and moisture is formed on the track such that the ball comes to a rest after 10 oscillations.

On the axes of Fig. 2.2, sketch the variation with time t of the displacement x of the ball, for the first two periods after moisture has formed. [2]

(iii) After many oscillations, the amplitude of oscillation decreased from 5.0 cm to 2.5 cm.

Calculate the loss in energy of the oscillation.

loss in energy = ... J [2]

[Total: 10]